

Introduction to Probability and Applied Statistics

Week 1: Data and measurement

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Welcome to GOVT 6019

Introduction to Probability and Applied Statistics

Plan for today: **course mechanics**, then **data and measurement**.

The teaching team



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What the course is for

- Critical **consumers** of empirical research: what a table says, and what had to be true for it to say that.
- Rigorous **practitioners**: cleaning, wrangling, and analyzing real political data in R.
- Clear **communicators** of results, in prose, tables, and figures.

Learning outcomes

- Summarize, visualize, and describe political data, and recognize what a given measure can and cannot support.
- Clean, manage, and analyze real datasets in R, and reproduce the analyses presented here.
- Use probability and random variables to represent a data-generating process.
- Assess an estimator's bias, variance, and consistency, and state precisely what a standard error, a confidence interval, and a test mean.
- Derive, estimate, and interpret the linear model, and state the assumptions its interpretation requires.

The two meetings

SEMINAR

Fridays, 2:00–4:30pm.

DISCUSSION

Wednesdays, 12:20–1:10pm.

Discussion meets on Wednesday and seminar on Friday, so discussion covers the **previous week's** seminar material.

Homework and exams

Homework is weekly and **ungraded**. Solutions go out the following week.

Component	Weight	When
Midterm	50%	Week 7, in class
Final	50%	Exam period, in person

Texts

- **Gailmard**, *Statistical Modeling and Inference for Social Science*.
- **Aronow & Miller**, *Foundations of Agnostic Statistics*. Where the derivations come from.
- **R4DS**, Wickham, Çetinkaya-Rundel & Grolemund: <https://r4ds.hadley.nz>
- **Model to Meaning**, Arel-Bundock, Parts I–II: <https://marginaleffects.com>
- **Healy**, *Data Visualization*: <https://socviz.co>

The last three are free online.

R with RStudio, both free: <https://posit.co/download/rstudio-desktop>

An early discussion section walks through installation and setup.

Why R:

- Free and open source
- A statistical programming language, plus packages for most methods
- Graphics
- Data handling
- Many free tutorials and reference books
- Steeper start than point-and-click software, larger payoff later



Getting help

Resources available to you:

- For homework and computing questions: Wednesday discussion section, TA office hours, your peers.
- For questions about lecture content: Wednesday discussion section, TA or instructor office hours, your peers.
- For course mechanics: check the syllabus first.

Work **together** on homework. Find your group or buddy today, and make a plan to meet weekly.

Both exams are individual, but preparation should be collaborative.



<https://govt6019.github.io/website/>

Devices in class

Bring a laptop to discussion section.

When we are not coding, close it. Phones silenced and put away. Slides go up on the course website before class, so you can take notes on paper.*



*If you have concerns about how this policy might affect your learning, please speak to me outside of class.

Accommodations and access

If you are registered with Student Disability Services, send your accommodation letter **in the first two weeks** where possible, so we have time to arrange things. Later is still fine.

Hit an access barrier in this course, printed, online, or in the room? Tell me or your SDS counselor right away.

SDS: <https://sds.cornell.edu> · 607-254-4545

If you are sick

Stay home. **You will not be penalized for missing class when you are sick,** provided you tell the teaching staff as soon as you can.

Catching up is yours to do: materials are on the course website, and get notes from a classmate. If you must come in while sick, or you are not sure, wear a mask. Ask us and we can usually supply one.

The AI policy

Exams: no AI tools, of any kind.

Homework and in-class coding practice: use it as a learning and coding aid. Debug an error message, get a concept explained a second way.

Homework is ungraded, so none of this is about credit. Work you outsource to AI is practice you did not get, and the midterm is where you find that out.

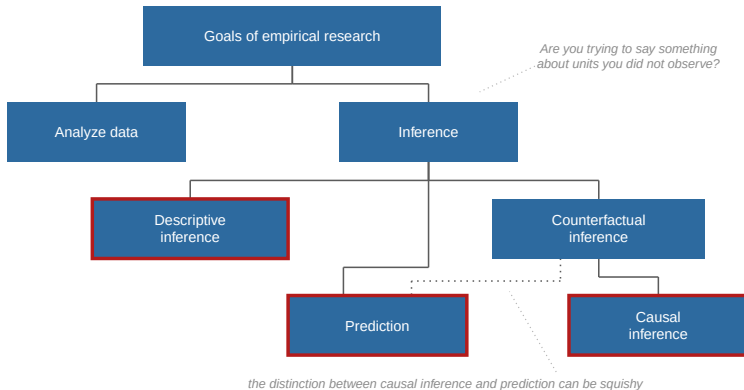
Unsure whether a particular use is allowed? Ask before you do it.

Course outline

Unit	Wk	What we cover
1 · Data	1–2	Data and measurement; describing empirical data
2 · Probability	3–4	Probability and data-generating processes; expectation
3 · Estimation and inference	5–6	Estimators, large samples, standard errors, tests
4 · Linear regression	8–12	OLS and its assumptions; extensions; inference; presenting results

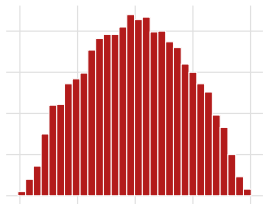
Dates are on the syllabus.

Goals of empirical research

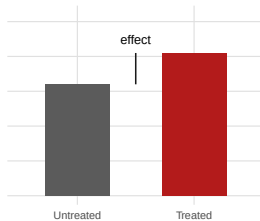


Three targets of inference

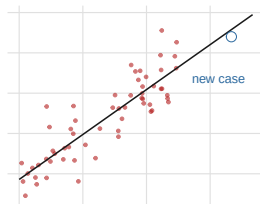
Descriptive
What is the case?



Causal
What would X change?



Predictive
What comes next?



Three targets of inference

DESCRIPTIVE

What is the case?

- Ideal points from roll-call votes (NOMINATE)
- Topics from speeches and texts
- Economic development from satellite images

CAUSAL

What would X change?

- Effect of a policy on turnout
- Effect of social media use on polarization
- Effect of severe weather on climate beliefs

PREDICTIVE

What comes next?

- Election outcomes from polls
- State failure from past instability
- Individual behavior from aggregate data

Roadmap

Data

Measurement

Notation

Wrapping up

Data vs. information

Data: Abstractions, representations, symbols, values describing qualities or quantities. → E.g. “All” “the” “world’s” “a” “stage”

Information: Signals that contain something systematic (not just noise) → E.g. “All the world’s a stage”

Data is a representation of underlying information.

Raw data may or may not contain interpretable information. Invariably, we need to *do things to and with data* to extract signal and information. *This is what much of data science is about.*

Numerical vs. quantitative vs. qualitative

Pretty much everything we do in this course will relate to **numerical** data:
Data represented as numbers.

But this does not mean that everything is **quantitative**.

Quantitative: Capturing a *quantity* as a continuous or discrete variable

Qualitative: Capturing a *quality* as a categorical variable

For our purposes, almost all data will need to be converted into **numerical** data, if it's not already.

Units and observations

In rectangular (tabular) data, rows are observations.

Units: The entities or subjects being studied (e.g., individuals, countries, companies).

Observations: A single “peek” at a unit under specific conditions (e.g., time period).

- Sometimes units = observations → *cross-sectional* datasets:
 - Take a slice (cross-section) at a single point in time
 - Can be done repeatedly (‘repeated cross-sections’)
- Often units $\neq$ observations → *hierarchical* datasets:
 - Each unit can have multiple observations (e.g., repeated measures, time series)
 - Can be done across different units (e.g., countries, individuals)

In GOVT 6019 our focus is on cross-sectional data.

Datasets

senator	party	state	born
Sanders	Ind.	VT	1941
Collins	Rep.	ME	1952
Warnock	Dem.	GA	1969

Rows are observations (your “unit of analysis”); **columns are variables**.

First question of any file: **what is a row?**

The unit of analysis

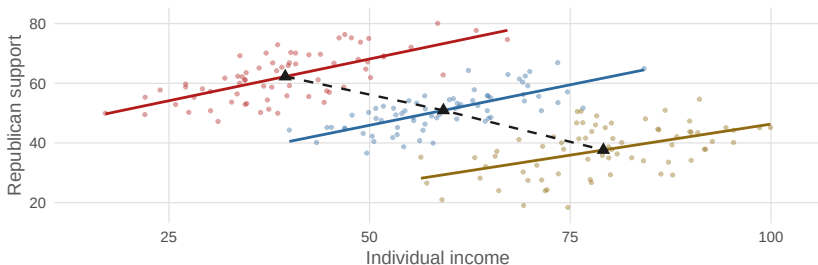
A single concept (e.g. “turnout”) can be measured at different units:

Unit of analysis	One row is	“Turnout” becomes
Individual	a registered voter	voted, 0/1
Precinct	a precinct	share of registrants voting
State	a state	statewide rate
Country-year	a national election	national rate

Which unit does your claim need?

The ecological fallacy (simulated)

Three hypothetical states, with the three state averages joined



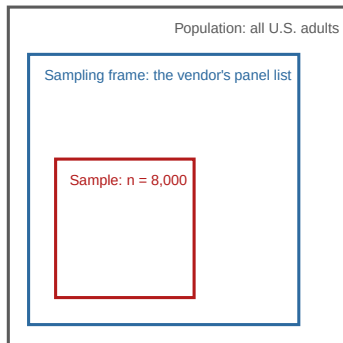
Within states, higher income **individuals** are **more** likely to vote Republican.

Across states, higher income **states** are **less** likely to vote Republican.

Reading an aggregate pattern as a claim about individuals is the **ecological fallacy** (Robinson 1950).

A reversal like this one is *Simpson's paradox* (Gailmard, section 2.2.1).

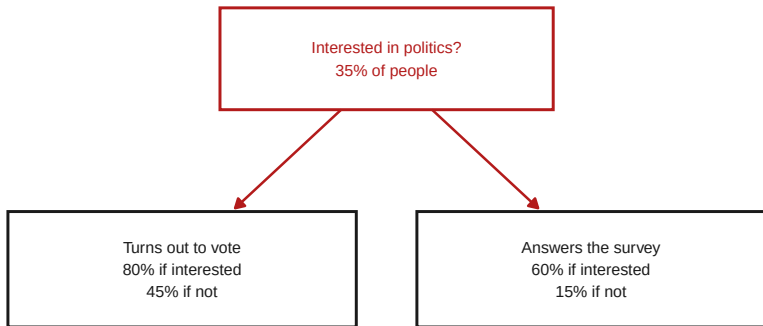
Population vs. sample



The **population** is the set of units the claim is about, and the **sample** is the rows you get.

Survey data: in between sits the **sampling frame**, the list you can actually draw from.

Sampling bias?



Interest in politics affects turnout, and it affects whether someone responds.

Sampling bias

```
N      <- 5000000                                # huge survey
interest <- rbinom(N, 1, 0.35)                    # interested?
y      <- rbinom(N, 1, ifelse(interest == 1, .80, .45)) # turnout
resp   <- rbinom(N, 1, ifelse(interest == 1, .60, .15)) # who answers

mean(y)                                           # population 57.2%
mean(y[resp == 1])                               # respondents 68.9%
```



This is assuming people report whether they vote truthfully. We know that they sometimes don't. That's another type of bias we worry about in survey research.

Where those numbers come from

	interested	not interested	total
Share of people	0.350	0.650	1.000
Chance of answering	0.60	0.15	
Share who answer	0.2100	0.0975	0.3075
	0.350(0.60)	0.650(0.15)	0.2100 + 0.0975
Share of respondents	0.683	0.317	1.000
	0.2100/0.3075	0.0975/0.3075	

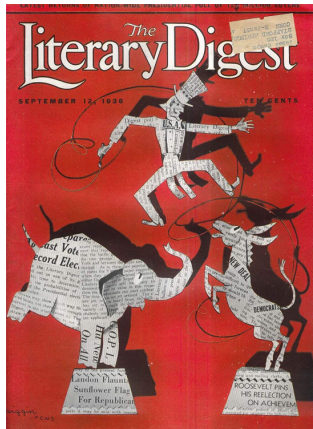
Turnout is then the two rates **0.80** and **0.45**, weighted by the shares in that row:

Population	$0.350(0.80) + 0.650(0.45)$	= 57.2%
Respondents	$0.683(0.80) + 0.317(0.45)$	= 68.9%

Election prediction goes wrong

The *Literary Digest* predicts Landon over FDR

- Poll predicts Alfred Landon victory in 1936 presidential election
- Reality: FDR carries 46 states
- Huge sample: 10 million polled; 2.4 million responded
- Non-random sample: magazine subscribers, automobile owners
 - Wealthy, Republican
- Gallup correctly predicted election results ($n = 50,000$)
 - Era of modern scientific public opinion research begins



Election prediction goes wrong, again

“Dewey Defeats Truman”

- Polls and early election returns suggested Dewey victory in 1948 presidential election
- Truman won 303–189 electoral college votes
- Polling stopped in mid-October; Truman surged late
- Quota sampling with interviewer discretion also led to unrepresentative sample



Getting to know your dataset

`voter2012.csv`: 8,000 U.S. adults, nationally representative 2012 survey;
9 variables in our subset.

important_econ	unimportant_budget	educ	birthyr	pol_interest	pid3
1	4	5	1931	3	1
1	4	3	1952	3	2
1	3	2	1931	3	1
1	4	3	1959	3	3

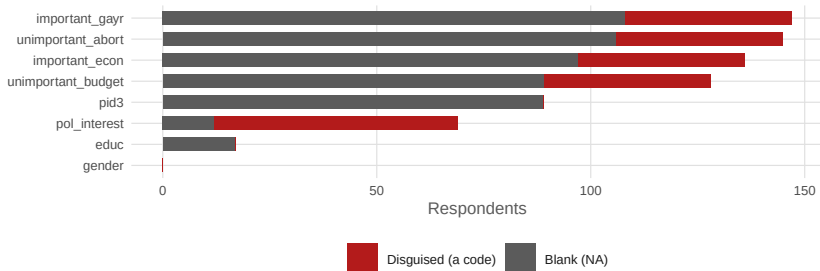
The codebook

Column	Codebook entry
educ	1 No HS · 2 HS · 3 Some college · 4 2-year · 5 4-year · 6 Post-grad
pol_interest	1 Not much · 2 Somewhat · 3 Very much · 4 Not sure
important_econ	"The economy is an important issue": 1 Strongly agree · · · 4 Strongly disagree · 5 Don't know

Rule: **codebook open before data open.**

Missing, and disguised as data

Item nonresponse in voter2012.csv, by column



How are missing values coded (or not coded) in your data?

First contact with a file

```
dim(d)           # rows and columns
glimpse(d)       # name, type, first values
colSums(is.na(d)) # how many NA in each column
map(d, \(x) sort(unique(x))) # every distinct value
map_dbl(d, \(x) max(x, na.rm = TRUE)) # the largest value, per column
```

Always take the time to look at your data before anything else.

`\(x)` is R's shorthand for `function(x)`, added in R 4.1. `map()` walks a data frame column by column and applies that function to each. `na.rm = TRUE` drops missing values before applying the function.

First contact with a file

```
dim(voter_raw)
```

```
[1] 8000    9
```

```
d <- voter_raw |> select(important_econ, pol_interest, educ)
dim(d)
```

```
[1] 8000    3
```

```
glimpse(d)
```

```
Rows: 8,000
```

```
Columns: 3
```

```
$ important_econ <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, ~
```

```
$ pol_interest   <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 2, 3, 3, 3, 3, 3, 3, 3, ~
```

```
$ educ           <dbl> 5, 3, 2, 3, 5, 4, 6, 2, 5, 3, 6, 5, 3, 6, 6, 5, ~
```

```
colSums(is.na(d))
```

important_econ	pol_interest	educ
97	12	17

First contact with a file

```
map(d, \(x) sort(unique(x)))
```

```
$important_econ
```

```
[1] 1 2 3 4 5
```

```
$pol_interest
```

```
[1] 1 2 3 4
```

```
$educ
```

```
[1] 1 2 3 4 5 6
```

```
map_dbl(d, \(x) max(x, na.rm = TRUE))
```

```
important_econ
```

```
5
```

```
pol_interest
```

```
4
```

```
educ
```

```
6
```

Roadmap

Data

Measurement

Notation

Wrapping up

From concept to score

Step	Choice	Example: political interest
Concept	the idea	attentiveness to political life
Conceptualization	which attributes	following news? caring?
Operationalization	the scoring rule	"How interested...?", scored 1-3
Scores	the data column	<code>pol_interest</code>

Each rung is a contestable choice (Adcock & Collier 2001).

Democracy, three ways

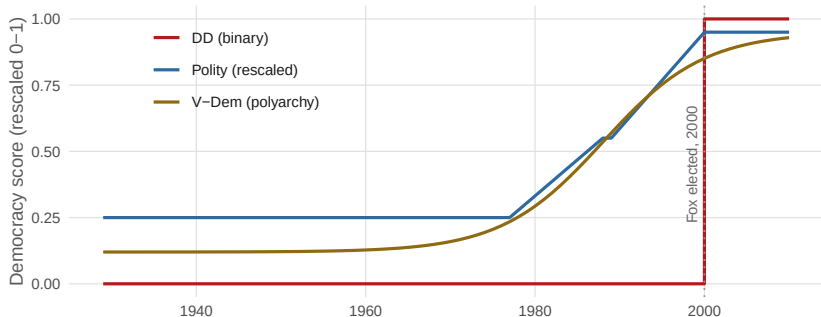
Measure	Democracy taken to be	Score
Polity	executive constraints, competitiveness	integer, −10 to 10
V-Dem	Dahl's polyarchy, aggregated expert ratings	continuous, 0–1
DD*	contested elections incumbents can lose	binary

Three defensible conceptualizations of the same concept.

*DD is the Democracy–Dictatorship measure of Przeworski et al. (2000), extended by Cheibub, Gandhi and Vreeland (2010).

Mexico, 1929–2010, under all three

Stylized paths



Levels of measurement

STEVENS (1946)

Nominal: unordered categories (party, region).

Ordinal: ordered, gaps uneven or unknown (education bands, agree–disagree).

Interval: equal gaps, but zero is just a point on the scale (birth year).

Ratio: equal gaps, and zero means *none of the thing*, so “twice as much” is a claim you can make (age, vote share).

Categorical covers nominal and ordinal. **Numeric** covers interval and ratio. **Binary** is nominal with two categories, and its mean is a share when it is coded 0/1.

A numeric variable is also either **discrete**, taking values you can count with nothing in between (seats won, children), or **continuous**, taking any value in a range (vote share, income).

The level is a property of the measure you chose, not of the concept.

Numeric and categorical

When you open a dataset, a categorical variable may be stored as either words or numbers. It will almost always require transformation (recoding) before it can be used in quantitative analysis. In `voter2012.csv`, `pid3` is party identification:

Code	1	2	3
Category	Democrat	Republican	Other

Caution: `mean(pid3)` returns a number, but that number is not meaningful. You must transform it (usually, to a binary indicator) before you can do arithmetic on it. For example:

```
table(voter_raw$pid3)
```

```
  1    2    3  
3639 3091 1181
```

```
Democrat <- ifelse(voter_raw$pid3 == 1, 1, 0)  
table(Democrat)
```

```
Democrat  
  0    1  
4272 3639
```

Classify the VOTER columns

- educ: ordered, uneven gaps. **Ordinal**.
- pid3: numbers as labels. **Nominal**.
- birthyr: equal gaps, zero is a convention. **Interval**.
- gender: two categories. **Nominal**, binary.
- pol_interest: **Ordinal**, 1–3.

code	n
1	509
2	2359
3	5063
4	57
NA	12

What is code 4?

Not sure.

Measurement error

$$\text{recorded}_i = \text{truth}_i + \underbrace{b}_{\text{systematic}} + \underbrace{u_i}_{\text{random}}$$

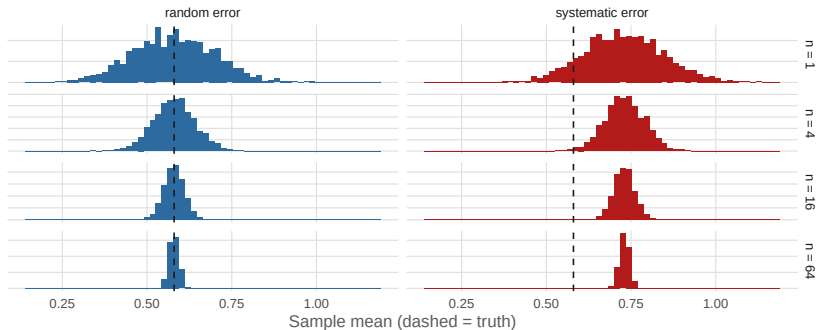
Random error u_i varies across units and averages to zero. **Systematic error** b is the same distortion for everyone.

Which one does a bigger sample fix?

Only **random error**. The systematic b survives any n .

Random and systematic error under aggregation

2,000 samples at each size; distribution of the sample mean



Both columns concentrate at the same rate. The right one is centred on the **wrong number**.

Systematic error in the field: turnout

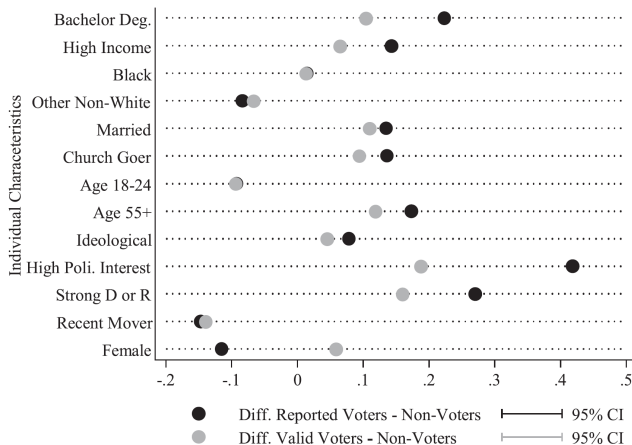
Say we want the correlates of turnout at the individual level.

Survey respondents misreport whether they voted. They say yes when the answer is no, because voting is what a good citizen does. The social pressure to say yes is strongest among the people already most likely to vote.

That is **systematic error**. A bigger survey will not remove it.

Ansolabehere and Hersh (2012) matched survey respondents to voter file records, so they can compare what people *said* to what they *did*.

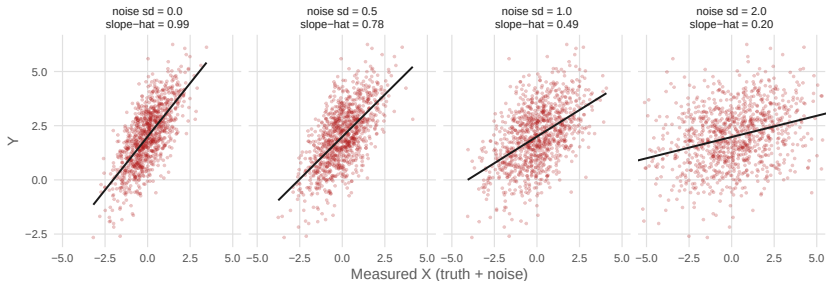
Who lies about voting



Ansolabehere & Hersh (2012), *Political Analysis* 20(4). Black: the voter/non-voter gap using self-reports. Grey: the same gap using validated votes.

What random error (noise) in X does to a slope

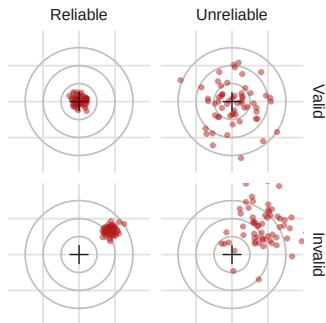
True slope is 1.0 in every panel



Noise in the *measure* of **X** flattens the estimated slope toward zero.

Reliability and validity

60 measurements; + = truth



Reliability is a tight cluster (small u_i). **Validity** is being on target (no b).

Reliability and validity, defined

DEFINITIONS

A measure is **reliable** if repeated application to the same unit returns the same score. It is **valid** if it scores the concept it claims to score.

Reliability is checkable from the data alone: test–retest, internal consistency, inter-coder agreement.

Validity is an argument, made against theory and against benchmarks.

Three kinds of validity

Face validity: does the measure *seem relevant* to the concept? (Scoring a prison inmate's likelihood of recidivism by their favorite color has none.)

Content validity: does it capture all the important facets of the concept? (Dating U.S. wars by formal congressional declaration misses every American war since 1945.)

Construct validity: does it line up with other measures of the same concept, from theory or previous research? (Mayhew counts a law as important only if it was judged so both at passage and in retrospect; the CRS list of "landmark laws" is retrospect alone, and the two do not agree.)

Which of these can a statistical test settle for you?

None of them (though in some cases it is possible to quantify construct validity; see Gailmard, section 2.1.3). Each one you have to argue for.

Gailmard, section 2.1.3.

Roadmap

Data

Measurement

Notation

Wrapping up

Notation

- Our notation will be consistent within this course, and it will often differ from what you meet in articles and textbooks.
- No one set of symbols covers every model, so the same letter gets reused across literatures for different things.
- Write down what each symbol means, in class and in your reading, and keep that list running.

Observations and subscripts

A column is a list, $X_1, X_2, \dots, X_n$, one entry per row, with n the number of observations.

The subscript is an address, and X_3 says whose value it is.

With $X = \text{birthyr}$: $X_1 = 1931$, $X_2 = 1952$, and so on, until the n th individual.

n is how many rows you have; N is how many the population has. Groups take a subscript, n_g , with $\sum_g n_g = n$.

Two exceptions you will meet in print: Gailmard writes N for the sample size, and regression tables report N for the number of observations.

$$\sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n$$

EXAMPLE

Ages 23, 27, 34: $\sum_{i=1}^3 X_i = 84$.

The mean divides that by n : $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 84/3 = 28$.

The index is bound: $\sum_j X_j$ is the same number.

Summation rules

With $X = (3, 5, 8)$, $Y = (1, 2, 4)$, $c = 2$:

Rule	Statement	Check
Constants	$\sum_{i=1}^n c = nc$	$2 + 2 + 2 = 6$
Scaling	$\sum cX_i = c \sum X_i$	$6 + 10 + 16 = 32$
Additivity	$\sum (X_i + Y_i) = \sum X_i + \sum Y_i$	$4 + 7 + 12 = 23$

$\sum$ does **not** pass through products or squares:

$$\sum X_i^2 = 98 \neq (\sum X_i)^2 = 256.$$

Indicators

$$\mathbf{1}\{A\} = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{otherwise} \end{cases} \quad \Rightarrow \quad \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{A_i\} = \text{share where } A \text{ holds}$$

```
grad <- as.numeric(voter_raw$educ >= 5) # 1 if 4-year degree or more
table(grad)
```

```
grad
  0    1
4801 3182
```

```
mean(grad, na.rm = TRUE) # the share, since the codes are 0/1
```

```
[1] 0.398597
```

Parameters and statistics

PARAMETERS

Features of the world. Fixed, unobserved.

Greek letters: μ, σ, β .

STATISTICS

Computed from n rows. Random before you see the sample.

Latin letters: $\bar{X}, \mathbf{s}, \mathbf{b}$; hatted, $\hat{\beta}$.

Estimand, estimator, estimate

THREE OBJECTS

The **estimand** θ is the quantity you want, a feature of the population. The **estimator** $\hat{\theta}$ is a rule that turns a sample into a number. The **estimate** is that number, for the sample in hand.

Here $\theta = \mu$, $\hat{\theta} = \bar{X}$, and the estimate is 53.3.

The estimator is a function of the data. What does that make it, before you see the sample?

A **random variable**: it has a distribution before you draw the sample.

Roadmap

Data

Measurement

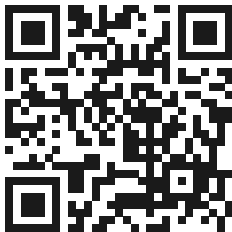
Notation

Wrapping up

Further reading

- Gailmard, *Statistical Modeling and Inference*, ch. 1 and section 2.1.
- Aronow & Miller, *Foundations of Agnostic Statistics*, Introduction.
- Adcock & Collier (2001), "Measurement Validity," *APSR* 95(3).
- Gelman et al. (2008), *Red State, Blue State, Rich State, Poor State*.

Background survey



<https://forms.gle/DqZ7pmuvyE5qtW8a6>